

Your Palm Is in My Hand: Palm-Vein Biometric Spoofing via Electromagnetic Side Channel

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Abstract

As a biometric modality, palm-vein authentication has seen growing adoption in recent years and is becoming prevalent in wallet-free payments and critical access-control applications. Although palm-vein patterns are widely regarded as nearly impossible to forge or steal, our study reveals that the electromagnetic (EM) emanations induced during sensor operation can leak information correlated with a user's palm-vein characteristics and attackers can use it to reconstruct palm-vein images and spoof authentication systems. In this paper, we present PVSpy, a novel attack framework that demonstrates the feasibility of this attack. Specifically, PVSpy passively captures EM emanations radiated from palm-vein sensors, extracts features that map the captured signals to palm-vein images, and exploits diffusion models for biometric reconstructions to spoof palm-vein authentication devices. We extensively evaluate PVSpy on five commercial devices, and the recovered vein patterns exhibit a promising level of similarity. In addition, we conduct 30 end-to-end spoofing attacks against these commercial-off-the-shelf (COTS) devices with different types of palm-vein sensors, and PVSpy achieves 26.7% (top-1) and 63.3% (top-3) success rates.

CCS Concepts

• Security and privacy → Side-channel analysis and counter-measures.

Keywords

Electromagnetic side channels; Palm vein authentication; Variational autoencoder

ACM Reference Format:

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Figure 1: Palm-vein authentication examples in the public.

1 Introduction

As an emerging biometric authentication paradigm, palm-vein recognition has gained significant attention in recent years. Its principal advantages of security, robustness, and convenience position it as an ideal solution for authentication in complex environments, particularly in public settings (Figure 1). Consequently, high-profile corporations and government departments have rapidly adopted palm-vein authentication for high-stakes identity verification and access control [3].

Palm-vein authentication possesses distinct security advantages over external biometric modalities, such as fingerprint, iris, and face. Since vascular patterns are subcutaneous, they are practically infeasible to capture from a distance without the user's explicit cooperation. This characteristic provides inherent and robust resistance against surreptitious capture and replay attacks. Furthermore, the palm-vein pattern is robust, feature-rich, and possesses high entropy. It remains unaffected by superficial issues such as skin dryness or makeup, and its internal nature makes it exceptionally resilient to brute-force or forgery attacks [15, 27, 34, 35]. In terms of usability, palm-vein recognition requires neither direct contact, as with fingerprint sensors, nor the precise alignment needed for face or iris-based systems. Instead, users need only hover their palm roughly above the palm-vein scanner. This contactless operation enhances user-friendliness and addresses critical hygiene concerns, which is particularly valuable for public interactions, as highlighted during the pandemic.

Despite the inherent security advantages of palm-vein biometrics, our research uncovers a critical vulnerability: COTS scanners can inadvertently leak sensitive information about users' palm-vein patterns. Typically, these scanners emit near-infrared (NIR) light onto the palm and capture the reflections using an image sensor. Since veins absorb more NIR light than the surrounding tissues, the sensor captures a distinct vascular pattern for authentication. However, during this process, the image sensor generates detectable EM emanations. These emanations correspond to current fluctuations caused by the varying light intensities of the reflected image. We demonstrate that these signals can be captured by low-cost hardware (e.g., a simple coil) without physical tampering. Our investigation shows that these leaked signals can be exploited to reconstruct high-quality palm-vein images. More severely, these reconstructed

images can be used to spoof the palm-vein authentication system, thereby enabling unauthorized access.

In this paper, we present PVSpy, a novel framework that demonstrates the feasibility of EM side-channel attacks against palm-vein scanners. To the best of our knowledge, this is the first study to reveal and physically validate the electromagnetic side-channel vulnerability in palm-vein authentication systems. In practice, the attack can be carried out with only low-cost equipment and a personal computer, underscoring the severity of this risk. In particular, PVSpy first captures the EM emanations from a scanner and applies filtering to the raw signals. Upon detecting the occurrence of an authentication event, it records and segments the raw EM signals containing the user's palm-vein data, and utilizes a denoising autoencoder to extract the palm-vein-related feature map. Then, PVSpy uses a pre-trained VQ-VAE-2 model to reconstruct the user's palm-vein image from this feature map. To further enhance the quality of the reconstructed image, a diffusion-based denoising module is implemented to remove noise that blurs the output. Finally, the reconstructed palm-vein image is used to spoof the palm-vein authentication system.

We conducted an extensive evaluation of PVSpy on five different COTS palm-vein scanners under different experimental settings. We developed a low-cost attack prototype based on an Arduino Uno R3 and commercial coils for EM emanation capturing. Our experimental results showed that the palm-vein images reconstructed by PVSpy achieved very high similarity to the originals across different scanners, various distances, and diverse coil selections. In addition, we conducted 30 end-to-end attack trials, achieving success rates of 26.7% for a single attempt (top-1) and 63.3% within three attempts (top-3). Moreover, we validated through evaluation that PVSpy could spoof authentication systems registered with real palms, underscoring the practicability and realistic significance of our design. Finally, we analyzed the software of COTS palm-vein devices, discussing security assumptions and potential risks, and discussed possible mitigating countermeasures.

Contributions. We summarize our contributions as follows:

- **A novel side-channel attack.** We present a novel side-channel attack against palm-vein scanners. By exploiting analog EM emanations produced during authentication, our method reconstructs palm-vein images and uses them to spoof the system. To the best of our knowledge, this is the first study to uncover and experimentally validate an analog EM side-channel vulnerability in palm-vein authentication systems.
- **An end-to-end attack framework.** We introduce PVSpy, an end-to-end attack framework that demonstrates the feasibility of this new side-channel attack. By leveraging EM emanations emitted during the authentication process, it reconstructs high-quality palm-vein images that can spoof near-infrared sensors and bypass palm-vein authentication systems, without requiring any target-specific prior knowledge.
- **Empirical evaluation.** We evaluated PVSpy's performance on five COTS palm-vein scanners under different impact factors. Our results show that it can effectively recover palm-vein images across different devices and successfully spoof palm-vein authentication systems.

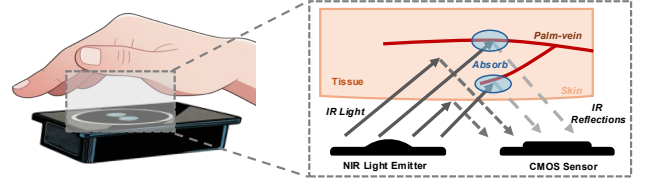


Figure 2: Illustration of a palm-vein scanner. A light source emits infrared illumination at specific wavelengths. As this light interacts with different human tissues, it is absorbed and reflected to varying degrees. A CMOS sensor captures the returning signal to produce the palm-vein image.

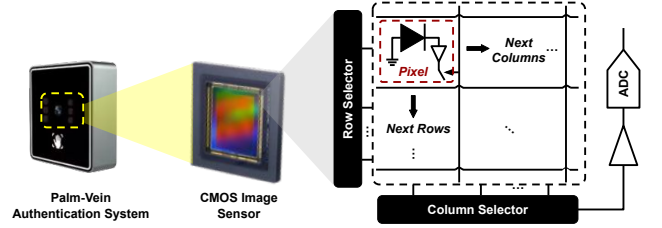


Figure 3: Illustration of principles of CMOS sensors deployed in palm-vein scanners.

2 Preliminaries

2.1 Palm-vein Authentication

Palm-vein recognition is a biometric modality that utilizes distinctive vascular patterns located beneath the skin. As illustrated in Figure 2, this technique relies on the principle that hemoglobin in blood vessels absorbs NIR light more strongly than the surrounding tissues. Consequently, veins appear as dark structures in the captured image. In contrast, the adjacent tissue contains less hemoglobin and reflects the light to appear brighter. Specifically, the device emits NIR light onto the palm, and a sensor equipped with a band-pass filter captures the reflections. The system subsequently extracts discriminative features from these images and compares them with enrolled templates for verification. This method offers high security and usability because the internal vein structure is difficult to image covertly. Furthermore, it is more robust to environmental factors than fingerprint or facial recognition systems. By requiring users to deliberately present their hands, the system also mitigates the uncertainties often associated with other contactless biometrics.

2.2 CMOS Sensor

Complementary Metal-Oxide-Semiconductor (CMOS) image sensors form the foundation of modern digital imaging. Known as active-pixel sensors, their fundamental architecture integrates photodetectors and amplification transistors directly within each pixel. This design allows essential signal processing functions like charge-to-voltage conversion and noise reduction to occur on the chip itself [9]. Compared to earlier technologies, this on-chip integration provides significant advantages including lower power consumption and faster readout speeds. Consequently, CMOS sensors now dominate a wide range of applications from consumer smartphones to industrial machine vision.

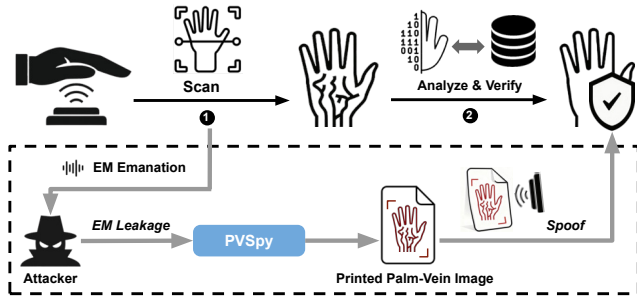


Figure 4: A motivating example: a user raises his/her hand and hovers above the public palm-vein scanner for authentication. The attacker can secretly capture the EM emanations during the scanning process, recover the user’s palm-vein image, and conduct a spoofing attack by presenting the printed or reconstructed image.

A key feature relevant to our study is the rolling shutter mechanism. Unlike global shutters that expose all pixels simultaneously, rolling shutters process images sequentially, line by line. As illustrated in Figure 3, during the readout phase, each pixel’s analog value passes through an amplification stage to an Analog-to-Digital Converter for digitization. This sequential process generates a time-varying current on the power supply lines that is directly modulated by the pixel intensities. Historically, side-channel research has primarily targeted robust digital signals from external interfaces. Internal analog signals from the CMOS readout are typically considered too weak and noisy to exploit. However, recent advances in computational power and machine learning now make it feasible to reconstruct information from these subtle analog channels. Our work leverages these advancements to align captured electromagnetic leakage with the raw sensor data.

3 Motivation and Threat Model

3.1 Motivating Example

Consider a typical payment scenario where a user presents their palm to a scanner. The device captures the vein pattern and verifies it against enrolled templates to authorize the transaction. However, an adversary can intercept the EM emanations generated during this process to reconstruct the biometric data and spoof the system. As illustrated in Figure 4, these EM emanations evolve across three authentication stages: (i) *Activation*: The device turns on its NIR light source and CMOS sensor, generating a distinct EM transient. (ii) *Sensing and Readout*: The sensor captures the palm-vein image. The analog front-end and ADC sequentially process pixel charges and transmit the digital data to the SoC. This process induces current fluctuations that leak EM emanations strongly correlated with the vein pattern. (iii) *Processing and Matching*: The SoC extracts features and compares them with enrolled templates. The EM signals observed here stem from the SoC’s computational activity.

An adversary can monitor these ambient emanations to launch a targeted attack. First, the activation signature serves as a trigger to pinpoint the start of the recognition process. Second, the adversary isolates the data-carrying signals from ① and removes irrelevant noise using the signals from (i) and ②. Finally, the attacker reconstructs the palm-vein image to authorize fraudulent transactions. Furthermore, emanations from ② can be analyzed to infer

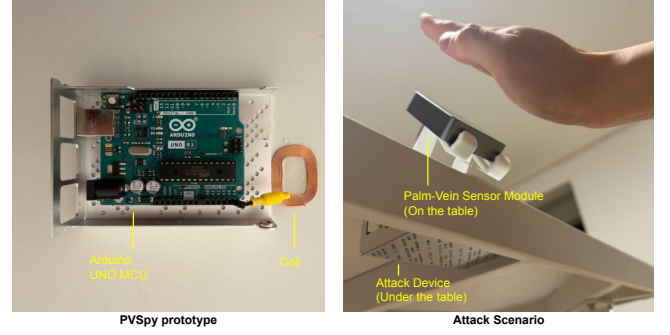


Figure 5: Illustration of a proof-of-concept attack scenario

the authentication outcome, thereby refining the attack strategy. This scenario highlights a critical threat: palm-vein biometrics are immutable and lifelong. Once compromised, they pose severe and irreversible security and privacy risks.

3.2 Threat Model

To demonstrate the attack’s feasibility, Figure 5 illustrates a proof-of-concept scenario simulating unattended deployments, such as payment terminals. A commercial palm-vein scanner is positioned on a desk, while a miniaturized interception device is concealed beneath the surface. This setup assumes the adversary has temporary physical access to covertly install the probe. Our threat model is strictly passive. The adversary cannot modify the target system’s hardware or software, and lacks visual line-of-sight to the victim’s palm. Instead, the attacker non-intrusively captures EM emanations during the scanning process to reconstruct the vein pattern and spoof the verifier. Consistent with prior EM side-channel research [8, 11, 22, 27, 28], we initially assume a short attack distance of under 10 cm.

Crucially, our prototype uses low-cost hardware to demonstrate a minimal barrier to entry. While this baseline relies on near-field interception, an advanced adversary equipped with commercial Software-Defined Radios (SDRs) could capture emanations from significantly greater distances. We explore this extended-range scenario further in the Discussion section.

4 Design of PVSpy

4.1 Overview of PVSpy

Figure 6 presents an overview of PVSpy. This framework exploits EM emanations emitted during the authentication process to reconstruct the user’s palm-vein pattern. The workflow proceeds as follows: (i) PVSpy uses a coil positioned near the sensor to continuously monitor and capture ambient EM signals. (ii) Next, the system detects authentication events using a sliding window approach. It isolates the signal segments containing palm-vein data by identifying characteristic EM features at the start and end of the scan. (iii) These isolated signals are processed to extract latent features corresponding to the palm-vein pattern. (iv) PVSpy employs a pre-trained neural network to reconstruct an initial palm-vein image from these features. (v) To improve image quality, a pre-trained denoising and upsampling network is applied. This step removes noise, enhances vein details, and increases the resolution to match the requirements of the target device. (vi) Finally, the

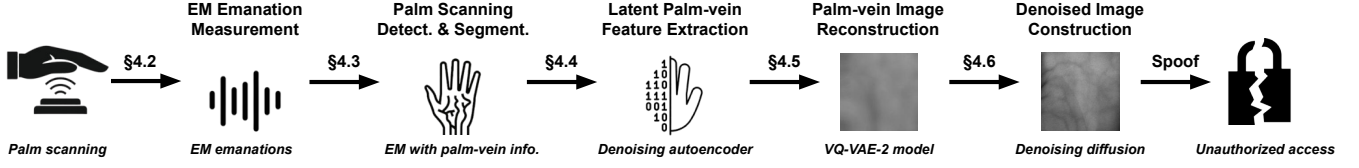


Figure 6: Overview of PVSPy

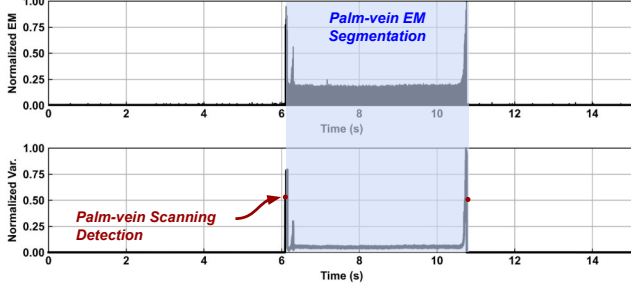


Figure 7: Palm-vein EM detection and segmentation.

reconstructed palm-vein image can be used to impersonate a legitimate user, enabling the adversary to bypass the authentication system for unauthorized payments or access.

4.2 EM Emission Measurements

To measure EM emanations during authentication, our threat model employs a compact, low-cost device equipped with a coil, positioned near the sensor. This setup is consistent with prior EM side-channel studies. In practice, palm-vein scanners are frequently deployed in unattended environments with minimal physical oversight. This allows an adversary to covertly install a small, passive receiver without attracting user attention. We implemented a prototype using an Arduino UNO R3 connected to a custom coil. This configuration demonstrates the attack’s feasibility in terms of cost and power efficiency. Although the Arduino board is used here for prototyping convenience, the essential components can be readily integrated into a coin-sized PCB or a thin film to maximize stealth. During operation, the module continuously captures EM signals and transmits them to the adversary.

4.3 Event Detection and Segmentation

Raw EM signals are inherently noisy due to environmental factors and interference from concurrently operating hardware. Additionally, since authentication is an intermittent event, the relevant signals are not continuous. Therefore, we preprocess the data to obtain a clean EM trace and isolate the segments containing palm-vein information. PVSPy continuously captures the raw signal and applies a Hampel filter to remove outliers, followed by a bilateral filter to suppress background noise. To pinpoint the authentication window, we apply a Dirichlet window and compute the wavelet variance to detect signal transitions. As illustrated in Figure 7, we identify two distinct change points in the variance profile to segment the specific interval containing palm-vein data. Finally, the extracted segment is normalized to the $[0, 1]$ range to eliminate amplitude variations and facilitate model training.

4.4 DAE Feature Extraction

A Denoising Autoencoder (DAE) [33] is an unsupervised neural network that extends a standard autoencoder by introducing noise

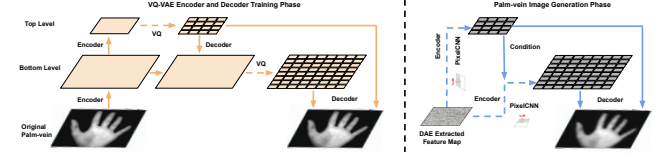


Figure 8: Architecture of the VQ-VAE-2 model.

to the inputs during training. This process forces the model to learn robust, noise-invariant feature representations without requiring labeled data. The architecture consists of an encoder, which compresses the input into a low-dimensional latent space, and a decoder that reconstructs the original signal. By minimizing the reconstruction loss between the clean target and the output generated from the noisy input, the model captures the essential signal structure while suppressing perturbations.

Following the isolation of EM signal segments containing palm-vein data, we employ a pre-trained DAE for feature extraction. Specifically, PVSPy partitions each segment into subsegments of length L and stacks them to form an input matrix of size $T \times L$. The DAE then maps this input to a latent representation of dimensions $i \times j$. In our implementation, we configured $L = f$ (the sampling frequency), $T = 3$, and $i = j = 64$. These parameters were empirically selected to strike a balance between capturing fine-grained vascular details and maintaining computational efficiency. Furthermore, the latent dimension was specifically chosen to align with the input resolution required by the subsequent VQ-VAE-2 model.

4.5 Palm-Vein Synthesis

To reconstruct the user’s palm-vein image from the extracted feature representations, we employ a modified Vector-Quantized VAE-2 (VQ-VAE-2) model [30]. This model maps the features derived from EM signals to palm-vein imagery. By employing a multi-level architecture, VQ-VAE-2 effectively addresses the blurriness typically associated with conventional autoencoders. This approach enhances detail preservation and suppresses artifacts, making it particularly suitable for restoring subcutaneous vascular patterns from imperfect EM measurements. The model excels at capturing complex morphological structures, such as vein bifurcations and microvascular networks, through progressive refinement across scales. Furthermore, its hierarchical latent representation disentangles global structure from local details, enabling the generation of high-fidelity anatomical features.

Figure 8 illustrates the VQ-VAE-2 architecture. We standardize the model input to single-channel 256×256 grayscale images. This resolution aligns with the effective Region of Interest (ROI) containing the primary vascular patterns in the majority of our test devices. Although some sensors support higher resolutions, normalizing inputs to 256×256 significantly reduces computational overhead without compromising recognition accuracy, as it preserves the core biometric features. These inputs are then compressed into

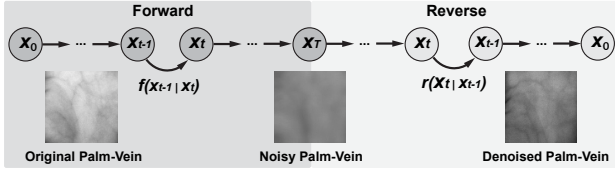


Figure 9: Illustration of the diffusion-based denoising model. Note that the noisy image x_T represents the reconstructed palm-vein from VQ-VAE-2. The reverse process is used to generate a denoised result.

quantized latent maps at two resolutions: a bottom-level map of 64×64 and a top-level map of 32×32 . This configuration is inherited from the original VQ-VAE-2, and was selected to strike a balance between preserving fine-grained vascular details and maintaining computational efficiency.

4.6 Diffusion based Denoising

Although VQ-VAE-2 can generate high-fidelity images, its reconstructions often contain residual noise. Such noise can obscure fine-grained palm-vein details and reduce similarity to the original biometric, which in turn lowers the success rate of spoofing attacks. To mitigate this issue, we add a denoising module based on Latent Diffusion Models (LDMs) [31]. The model learns a noise distribution and uses it to progressively refine the reconstructed images. As shown in Figure 9, the overall procedure consists of a forward noising process and a reverse denoising process.

Forward Process. The forward process gradually adds noise to the data. At each timestep, we perturb the original image with a small amount of Gaussian noise. As t increases, fine anatomical details fade until the sample becomes close to pure noise. Formally, let $x_0 \sim p(x)$ denote a recovered palm-vein image. We generate a noisy sample x_t at timestep t from x_{t-1} through $p(x_t|x_{t-1}) = \mathcal{P}(x_t, \text{noise}(t, t-1))$, where $\text{noise}(t, t-1)$ is the perturbation applied to x_{t-1} to obtain x_t . This forward process characterizes how noise accumulates over time and provides the noise statistics needed for the reverse denoising stage.

Reverse Process. The reverse process aims to reconstruct the original data by iteratively denoising the signal. Starting from a Gaussian noise distribution, the model progressively removes noise to restore image details. At each step, the network learns to recover a cleaner sample from the current noisy state. By leveraging the noise distribution learned during the forward process, the model reverse-infers the data distribution at each timestep to generate a clear image. Through multiple iterations, the model ultimately reconstructs a noise-free output. The reverse transition at timestep t is defined as: $q(\hat{x}_{t-1}|\hat{x}_t) = Q(\hat{x}_{t-1}; \text{denoise}(t-1, t))$, where $\text{denoise}(t-1, t)$ represents the denoising operation mapping $\hat{x}_t$ back to $\hat{x}_{t-1}$ using the deep learning model. By iteratively applying this process to the outputs of the VQ-VAE-2, we obtain a refined, high-quality palm-vein image.

5 Evaluation

5.1 Experimental Setup

5.1.1 Dataset. Palm-vein imagery represents highly sensitive biometric data. To mitigate privacy and security risks, we evaluate PVSpy exclusively using publicly available data, rather than collecting proprietary samples. Specifically, we utilize four palm vein

datasets: the Kaggle dataset [26], SCUT-PV-v1 [24], CASIA Palmprint Image Database [13], and PolyU-IITD Contactless Palmprint Images Database [18]. Consistent with the hardware specifications and effective ROI analysis discussed in §4.5, we extract a square crop from each raw image. This dimension was selected to fully encapsulate the central palmar vascular structure found in the majority of COTS scanners, while efficiently filtering out irrelevant background clutter. All imagery is intensity-normalized and aligned to minimize variance across subjects and sessions. We employ a subject-disjoint split, allocating 80 individuals for training and the remaining 18 for evaluation, ensuring no identity overlap between partitions. Where appropriate, we apply mild data augmentation to enhance robustness against acquisition variability without altering vascular topology.

5.1.2 Data Collection and Experimental Setup. To emulate realistic inputs to the CMOS palm-vein sensor, we present high-resolution printed palm-vein images to the device. Because these sensors capture 2D intensity images without depth information, a planar printout serves as an accurate proxy for the sensing stage. In practice, adversaries can exploit this vulnerability using direct printouts in unattended environments or by printing patterns onto latex gloves to mimic natural hand placement. To validate this approach, we conducted an IRB-approved pilot study involving five volunteers. We captured their live palm-vein images, printed them, and presented the printouts to the scanners. The systems successfully authenticated all five printed proxies. This confirms that our 2D proxies trigger the exact same authentication pipeline as live hands, validating the relevance of our subsequent EM measurements.

For each printed proxy, we conducted 60 authentication trials. During each trial, we recorded EM emanations at a 20 kHz sampling rate for 10 seconds, which is sufficient to capture relevant signal variations [17, 27]. This continuous 10-second window captures both ambient noise and the actual 3-second authentication event. Using this broad temporal window allows us to effectively evaluate the framework’s ability to autonomously detect and extract the target signal in a realistic passive attack scenario. To simulate natural hand placement, we introduced random planar rotations ($\pm 30^\circ$) and curvature ($\pm 10^\circ$) to the printouts. We assume standard, cooperative user behavior throughout the evaluation; adversarial behavior is out of scope.

All experiments were conducted on a workstation equipped with an AMD Ryzen 9 9950X CPU, 128 GB RAM, and dual NVIDIA RTX 5090 GPUs. We trained the Denoising Autoencoder, the modified VQ-VAE-2, and the Latent Diffusion Model using data from 80 subjects. To ensure a rigorous evaluation, we maintained a strict subject-disjoint split, reserving the remaining 18 subjects exclusively for testing the end-to-end reconstruction attack. Finally, we assessed the attack’s effectiveness using both image-level fidelity metrics and downstream biometric authentication success rates.

5.1.3 Evaluation Metrics. We use two algorithms to evaluate the similarity between the generated palm-vein image and the original palm-vein image: Perceptual Hash (pHash) and the Universal Quality Image Index (UQI).

pHash. Perceptual Hash is robust to palm-vein distortion and rotation [27], and can distribute two different palm-veins. We follow the previous work to select the pHash similarity as the evaluation

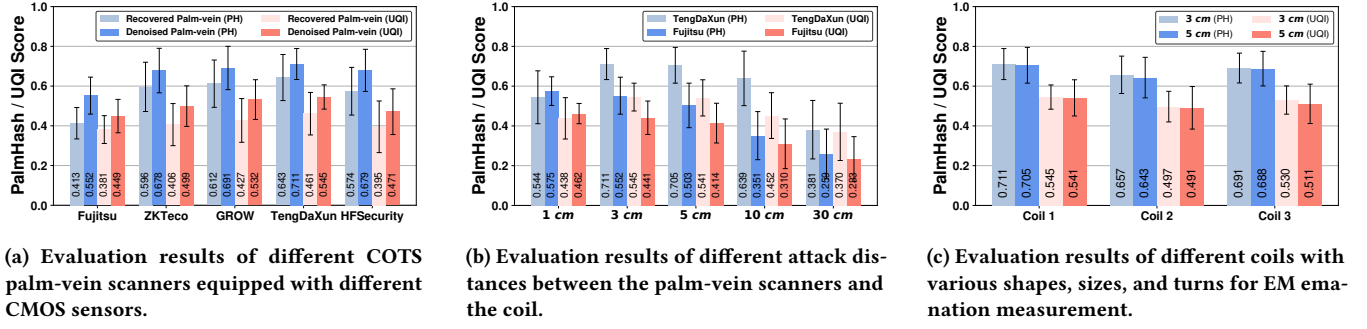


Figure 10: Impact of various physical parameters on the attack performance of PVSpy.

metric. We use f_r and f_o to denote the reconstructed palm-vein image and the original palm-vein image. It first uses a loop structure to apply the Discrete Cosine Transform (DCT) to calculate their pHash values by converting the palm-vein data streams from the spatial domain to the frequency domain. For each point $f_r(i, j) \in f_r$ and $f_o(i, j) \in f_o$, we calculate:

$$F_r = c(u)c(v) \sum_i \sum_j f_r(i, j) \cos\left[\frac{(i+1/2)\pi}{N_r} u\right] \cos\left[\frac{(j+1/2)\pi}{N_r} v\right],$$

$$F_o = c(u)c(v) \sum_i \sum_j f_o(i, j) \cos\left[\frac{(i+1/2)\pi}{N_o} u\right] \cos\left[\frac{(j+1/2)\pi}{N_o} v\right]$$

We obtain the upper left 8×8 pixels with high frequencies:

$$F_r^{doi} = F_r(u, v)[0 : 8, 0 : 8], F_o^{doi} = F_o(u, v)[0 : 8, 0 : 8]$$

Then obtain pHash values from F_r^{doi} and F_o^{doi} :

$$H_r = (F_r^{doi} > \text{average}(F_r^{doi})), H_o = (F_o^{doi} > \text{average}(F_o^{doi}))$$

At last, the algorithm calculates the hamming distance between two pHash sequences to obtain the similarity of the two palm-vein. **UQI**. The Universal Quality Image Index (UQI) is a metric designed to assess image quality by modeling distortion as a combination of loss of correlation, luminance distortion, and contrast distortion. It has a range from 0 to 1, where a UQI of 1 indicates high image quality, and lower values correspond to greater distortion. Only images showing the same scene should be compared, since UQI is straightforward to calculate and aligns well with human perception of image quality, making it an effective tool to evaluate visual fidelity.

We utilize UQI to calculate the quality index of the generated palm-vein image in comparison to the original palm-vein image. It models any image distortion as a combination of three factors: loss of correlation, luminance distortion, and contrast distortion. Let $x = \{x_i \mid i = 1, 2, \dots, N\}$ and $y = \{y_i \mid i = 1, 2, \dots, N\}$ be the original and the test image signals, respectively. The proposed quality index is defined as the product of three components:

$$Q = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \cdot \frac{2\bar{x}\bar{y}}{(\bar{x})^2 + (\bar{y})^2} \cdot \frac{2\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{4\sigma_{xy}\bar{x}\bar{y}}{(\sigma_x^2 + \sigma_y^2)[(\bar{x})^2 + (\bar{y})^2]}$$

where

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i, \sigma_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2,$$

$$\sigma_y^2 = \frac{1}{N-1} \sum_{i=1}^N (y_i - \bar{y})^2, \sigma_{xy} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}).$$

We use a sliding window to calculate the UQI of an image. In particular, a 8×8 window moves horizontally and vertically from the top-left corner, traversing all rows and columns of the image, calculating the local quality index Q_j at each step. If there are a total of M steps, then the overall quality index is given by: $Q = \frac{1}{M} \sum_{j=1}^M Q_j$.

5.2 Effectiveness

In our evaluation, we employ the pHash and UQI metrics to assess the generative performance of PVSpy and the effectiveness of each

system component. Specifically, we compare the ground-truth images against the intermediate reconstructions from the VQ-VAE-2 model and the final denoised images produced by the diffusion model. This comparative setup aligns with our threat model where the adversary cannot modify the hardware or software of the sensor and lacks access to internal raw data. We further investigate multiple practical factors that impact attack performance. The default attack distance is set to 3 cm. This configuration reflects common deployment scenarios where palm-vein modules are placed on a desktop or mounted on a standalone bracket. In these settings, a passive EM probe can be unobtrusively concealed beneath the desk surface or behind the mounting frame while remaining within the necessary near-field coupling range.

5.2.1 Different Palm-Vein Modules. Typically, different palm-vein modules integrate with distinct CMOS image sensors. Therefore, we selected five commercial palm-vein sensor modules to evaluate PVSpy across different devices, including Fujitsu PalmSecure Sensor V2 [4], ZKTeco PAR200 [5], GROW FP002 [1], TengDaXun PL-TM-P2 [6], and HFSecurity Palm Vein VP930 [2]. We separately collected data on each device to train different models for evaluations. As shown in Figure 10a, palm-vein images reconstructed by PVSpy on the aforementioned five palm-vein modules achieve pHash scores of 0.413, 0.596, 0.612, 0.643, and 0.574, and UQI scores of 0.381, 0.406, 0.427, 0.461, and 0.395, respectively. After applying the denoising diffusion model, the pHash results increase to 0.552, 0.678, 0.691, 0.711, and 0.679, with UQI values of 0.449, 0.499, 0.532, 0.545, and 0.470, respectively.

The results show that PVSpy is effective in attacking different COTS palm-vein modules, while the diffusion models substantially improve the performance. In particular, we notice that Fujitsu PalmSecure Sensor V2 presents lower performance than other modules, and the reason is that it contains additional RF shielding on the mainboard relative to the other devices, which reduces electromagnetic interference and concurrently attenuates EM emanation strength. Thus, we can enhance PVSpy's performance on this module by applying high-gain antennas or receivers. Nevertheless, PVSpy demonstrates the feasibility of reconstructing palm-vein information from EM emanations across different palm-vein authentication devices.

5.2.2 Different Attack Distance. According to Maxwell's equations, electromagnetic field strength decreases with distance, approximately following an inverse-square relationship. To quantify the effect of different distances on PVSpy, we use the TengDaXun PL-TM-P2 and the Fujitsu PalmSecure Sensor V2 as the baselines for

evaluation. Note that our default attack distance is 3 cm, and the four additional standoff distances are 1 cm, 5 cm, 10 cm, and 30 cm. We evaluate only the quality of the final generated images. As shown in Figure 10b, the results at 1 cm, 3 cm, 5 cm, 10 cm, and 30 cm are as follows: for TengDaXun PL-TM-P2, pHash scores are 0.544, 0.711, 0.705, 0.639, and 0.381, and UQI scores are 0.438, 0.545, 0.541, 0.452, and 0.370; for the Fujitsu PalmSecure Sensor V2, pHash scores are 0.575, 0.552, 0.503, 0.351, and 0.259, and UQI scores are 0.462, 0.441, 0.414, 0.310, and 0.233, respectively.

In particular, in TengDaXun PL-TM-P2 and Fujitsu PalmSecure Sensor V2, PVSpy performs better at 3 cm than 1 cm due to severe interference contaminating the EM emanations resulting from other internal components. When launching the attack in a long distance (e.g., Fujitsu PalmSecure Sensor V2: 10 cm, TengDaXun PL-TM-P2: 30 cm), the reconstruction quality of PVSpy degrades significantly due to the greater standoff leads to substantial attenuation and increased ambient noise, resulting in a considerable SNR decrease. In general, PVSpy demonstrates promising and stable effectiveness in reconstructing palm-vein biometrics on the two modules at a distance between 3 cm and 5 cm.

5.2.3 Different EM Coils. As shown in Figure 10c, we measure the impact of coil geometry on PVSpy’s attack performance. We use *coil*₁ (rectangular, 42 mm × 29 mm, 27 turns) as our baseline, which is also used in the preceding measurements, and we evaluate *coil*₂ (rectangular, 30 mm × 23 mm, 15 turns) and *coil*₃ (circular, diameter 42 mm, 10 turns) as additional configurations. We conducted evaluations on the TengDaXun PL-TM-P2 at attack distances of 3 cm and 5 cm. At the distance of 3 cm, the pHash scores of the three coils are 0.711, 0.657, and 0.691, with UQI scores of 0.545, 0.497, and 0.530, respectively. At the distance of 5 cm, the pHash scores of the three coils are 0.705, 0.643, and 0.688 with UQI scores of 0.541, 0.491, and 0.511, respectively. Overall, we found the size and shape of the coils to capture EM emanations present limited impact on PVSpy’s performance. This is because coil geometry primarily scales the amplitude of the captured EM emanations while offering little improvement in SNRs, and the signal processing normalizes the EM signals to mitigate the impact. Therefore, PVSpy can operate with different coils, capturing EM emanations and reliably reconstructing users’ palm-vein information.

6 End-to-End Attacks

6.1 Attack Setup

To further evaluate the effectiveness of PVSpy, we conducted 30 end-to-end physical attacks on the five COTS devices described earlier. For each device, we selected three left-hand and three right-hand palm-vein images from different individuals within the test set of 18 participants. We printed these ground-truth images to enroll them on the sensors and subsequently attempted to unlock the system using the reconstructed vein patterns generated by PVSpy. Each attack trial consisted of three unlocking attempts. We considered the trial successful if any one of the three attempts passed authentication. Based on this criterion, we computed the Top-1 and Top-3 success rates. We adopted the Top-3 metric because standard biometric systems typically permit a limited number of retries, ranging from three to five attempts, before falling back to alternative methods like passwords or smart cards. Assuming the model is pre-trained, the complete attack process requires approximately

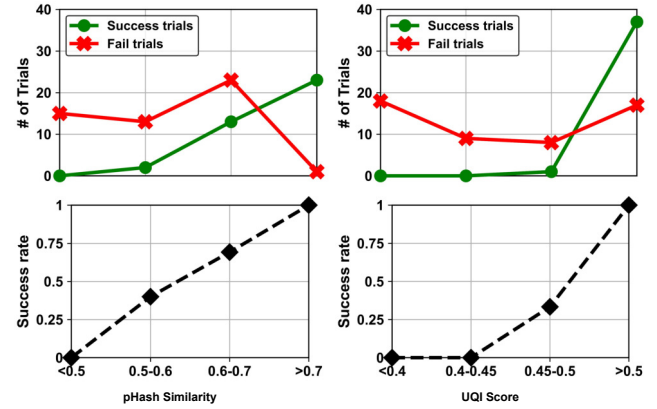


Figure 11: Top-3 success and fail attack trials, and ASRs with different ranges of PalmHash similarity and UQI score.

three minutes. This duration includes 5 to 10 seconds for EM signal capture, 2 to 3 minutes for image reconstruction, and 10 seconds to print the final attack image.

6.2 Attack Result

Table 1 summarizes the results of the 30 end-to-end attack trials, detailing the enrollment samples, reconstruction quality metrics (PalmHash and UQI), and success rates for Top-1 and Top-3 attempts. Aggregated across all devices, the Top-1 success rate is 26.7% (8/30), which increases to 63.3% (19/30) when allowing three attempts. Crucially, we observe a positive correlation between reconstruction quality and attack success. For instance, the Fujitsu PalmSecure Sensor V2 yielded lower quality scores (PalmHash 0.552, UQI 0.449) and a corresponding Top-3 success rate of 50%. In contrast, the TengDaXun VP910 exhibited higher reconstruction fidelity (PalmHash 0.711, UQI 0.545) and achieved a significantly higher success rate of 83.3%. These results confirm that higher similarity to the ground truth directly translates to higher spoofing success probabilities in practical scenarios. Furthermore, the data highlights the critical role of retry policies: reconstructions with marginal quality often fail on the first attempt but succeed within three tries. Collectively, these findings demonstrate the empirical effectiveness of PVSpy and underscore the necessity for system-level defenses, such as limiting retries, enhancing liveness detection, and improving EMI shielding.

6.3 Case Study

We analyzed the correlation between image quality metrics and attack success rates across 30 end-to-end trials. As illustrated in Figure 11, the results exhibit a clear dependency on reconstruction fidelity. For high-quality images with PalmHash exceeding 0.7 or UQI above 0.54, the Top-1 success rate was 77.78%, reaching 100% within three attempts. Conversely, attacks consistently failed when similarity scores were low (PalmHash below 0.5 or UQI below 0.45). The intermediate range (PalmHash 0.5–0.7, UQI 0.45–0.54) revealed a notable phenomenon: while the Top-1 success rate was only 6%, the Top-3 rate rose significantly to 64.7%. Crucially, among the successful cases in this group, the majority succeeded only on the final attempt. To investigate this behavior, we reverse-engineered the software driver of the HFSecurity device and identified specific verification logic that explains these results. We observed that the system extracts a broad feature set

Table 1: End-to-end attack results, including 30 trials of five COTS palm-vein scanners with different CMOS sensors introduced in §5.1.2. We randomly select recovered palm-vein images from both left and right hands in the testing set. Top-1: one attempt, Top-3: three attempts, “○”: spoofing denied, “●”: spoofing succeeded, passing authentication.

Trial	Device	Palm-vein Info.	PalmHash Similarity		UQI Score		Top-1 Results	Top-3 Results			Success?
								1st	2nd	3rd	
1	Fujitsu	No. 84, Left hand		0.561		0.501	○	○	○	●	✓
2		No. 83, Left hand		0.403		0.426	○	○	○	○	✗
3		No. 89, Left hand		0.397		0.361	○	○	○	○	✗
4		No. 90, Right hand		0.508		0.477	○	○	●	○	✓
5		No. 89, Right hand		0.381		0.311	○	○	○	○	✗
6		No. 94, Right hand		0.613		0.532	○	○	●	●	✓
7	ZKTeco	No. 91, Left hand		0.641		0.395	○	○	○	○	✗
8		No. 97, Left hand		0.739		0.566	●	●	●	●	✓
9		No. 93, Left hand		0.602		0.527	○	○	○	●	✓
10		No. 89, Right hand		0.692		0.544	○	○	●	●	✓
11		No. 88, Right hand		0.752		0.595	●	●	●	●	✓
12		No. 94, Right hand		0.561		0.439	○	○	○	○	✗
13	GROW	No. 91, Left hand *		0.699		0.423	○	○	○	○	✗
14		No. 85, Left hand *		0.724		0.550	●	●	●	●	✓
15		No. 88, Left hand		0.635		0.565	●	●	○	●	✓
16		No. 81, Right hand		0.324		0.289	○	○	○	○	✗
17		No. 82, Right hand		0.650		0.530	○	○	○	●	✓
18		No. 83, Right hand		0.521		0.461	○	○	○	○	✗
19	TengDaXun	No. 85, Left hand		0.799		0.601	●	●	●	●	✓
20		No. 87, Left hand *		0.613		0.494	○	○	○	○	✗
21		No. 82, Left hand		0.718		0.540	○	○	●	●	✓
22		No. 97, Right hand *		0.802		0.631	●	●	●	●	✓
23		No. 89, Right hand		0.641		0.511	○	○	○	●	✓
24		No. 84, Right hand		0.740		0.587	●	●	●	●	✓
25	HFSecurity	No. 96, Left hand		0.301		0.251	○	○	○	○	✗
26		No. 98, Left hand		0.681		0.505	○	○	○	●	✓
27		No. 89, Left hand		0.691		0.531	○	○	●	●	✓
28		No. 98, Right hand		0.742		0.581	●	●	●	●	✓
29		No. 91, Right hand		0.656		0.513	○	○	○	●	✓
30		No. 93, Right hand		0.570		0.384	○	○	○	○	✗

during enrollment but verifies against a randomly selected subset, a strategy likely intended to prevent replay attacks. Furthermore, dynamic analysis revealed an adaptive thresholding mechanism where high-similarity but failed attempts trigger a slight relaxation of the matching criteria for subsequent retries. We infer that this design aims to optimize user experience under the assumption that palm-vein patterns are practically infeasible to forge. This adaptive mechanism explains why reconstructions with marginal quality often fail initially but succeed on the third attempt. Similar design patterns were observed across the other four tested devices.

Figure 12 presents four representative ground truth images, their PVSpy reconstructions, and the corresponding PalmHash and UQI scores. We focus on anomalous cases that receive high scores but fail the authentication process. As highlighted by red rectangles, these reconstructions correctly localize vein positions but fail to reproduce precise vascular shapes. We attribute these structural errors to the limited training dataset size. Furthermore, these failed attempts often exhibit high PalmHash scores but low UQI values. This indicates that UQI is more sensitive than PalmHash to the fine textural details necessary for successful spoofing.

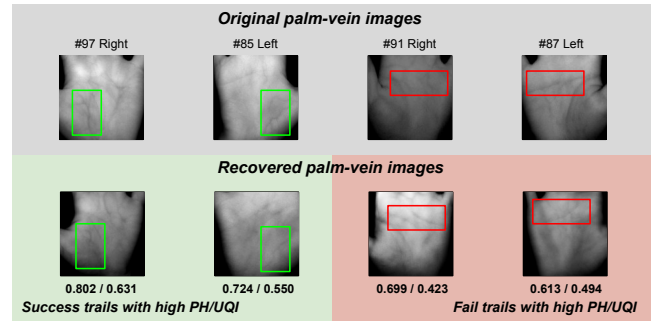


Figure 12: Illustration of denoised and original palm-veins in four end-to-end attack trials with high PalmHash/UQI score (marked *). Two of them successfully spoof the palm-vein scanner, while the others are not. Green/Red boxes: Similar/mispredicted patterns

6.4 Real-world Attacks

While our previous evaluation demonstrated the feasibility of PVSpy using public datasets, it is essential to validate its performance against live subjects in a real-world setting. To address this, we

Table 2: End-to-end attack results, including 25 trials of using recovered palm-vein images to spoof five different palm-vein scanners registered with real palm-veins. Top-3: three attempts, “○”: spoofing denied, “●”: spoofing succeeded.

Device	Device Resolution	Test Resolution	Top-3 of Five Real Palm-vein Images				
			PV-1	PV-2	PV-3	PV-4	PV-5
Fujitsu	(480, 640)	(256, 256)	○ ○ ○	○ ○ ○	○ ○ ○	○ ○ ○	○ ○ ○
ZKTeco	(480, 640)	(256, 256)	○ ○ ○	● ○ ○	○ ○ ○	○ ○ ○	○ ○ ○
GROW	(480, 640)	(256, 256)	○ ○ ○	○ ○ ○	○ ○ ○	○ ○ ●	○ ○ ○
TengDaXun	(1920, 1080)	(256, 256)	○ ○ ●	○ ○ ○	○ ○ ○	○ ● ○	○ ○ ●
HFSecurity	(1920, 1080)	(256, 256)	○ ○ ○	○ ○ ○	○ ○ ○	○ ○ ○	○ ○ ○

conducted an additional experiment involving five real palm-vein instances under strict IRB oversight. Given that palm-vein patterns are sensitive biometric data and generally permanent, testing on external participants presents specific ethical challenges regarding data privacy. We therefore minimized these risks by having the authors serve as the subjects for this specific experiment. The experimental procedure consisted of three steps. First, we enrolled our own palm veins into each authentication system. Second, we recorded the EM emanations produced during the standard verification process. Third, we reconstructed the palm-vein images using PVSPY and used them to attempt to bypass the scanners. All other experimental configurations remained identical to the protocols described in §6.

Table 2 summarizes the attack results and specifies the capture resolution for each device. While PVSPY successfully recovered palm-vein patterns in all cases, it bypassed the authentication system in 5 out of 25 trials (20%). We observe a performance attenuation compared to the results on the low-resolution public dataset presented in §6. We attribute this decline primarily to the resolution disparity: production devices capture images at significantly higher resolutions than the public datasets used to train our model, which hinders precise feature matching. Furthermore, the domain gap between live palm veins and printed proxies leads to different sensor responses. Crucially, however, given the immutability of biometric traits, this success rate represents a critical security threat. Unlike passwords, compromised palm-vein patterns cannot be revoked or reset; thus, even a limited number of successful breaches results in permanent exposure for the victim. Notably, the majority of successful attacks occurred on the final attempt, a finding that aligns with the adaptive thresholding mechanism identified in our reverse-engineering analysis. These results underscore the critical reliance on software-level controls and highlight vulnerabilities stemming from overly optimistic assumptions about hardware security. Finally, we acknowledge that the generative quality of PVSPY is currently constrained by the resolution of available training data. Overall, while PVSPY demonstrates strong attack potential, these findings indicate specific areas for refinement which we address in §7.2.

7 Discussion

7.1 Countermeasures

We formulated the following two directions of countermeasures and discussed them with vendors of palm-vein authentication systems, and provided practical solutions for mitigating the threat of our attacks.

EM Emanations Mitigation. To mitigate electromagnetic leakage at the hardware level, vendors must redesign sensor architectures [14] or implement physical shielding [28]. Alternatively, we propose using active signal modulation as a defense mechanism. While prior studies use electromagnetic interference offensively [16], systems can defensively inject dynamic noise during the scanning process to mask sensitive emanations. Without prior knowledge of these specific noise patterns, adversaries cannot practically extract the underlying biometric data.

Liveness Detection. Current palm vein systems rely on the flawed assumption that vascular patterns are secure against fabrication. Our experiments show that standard printers or electronic ink screens can bypass existing countermeasures like optical filters. Furthermore, discarding raw images after processing cannot prevent pattern reconstruction from electromagnetic leakage. To improve security, future systems could adopt dynamic blood flow analysis [7], 3D spatial sensing [29], or behavioral biometrics [32]. While these defenses mitigate physical spoofing, they fail to resolve the fundamental vulnerability of electromagnetic leakage.

7.2 Limitations and Future Works

PVSPY demonstrates the feasibility of reconstructing palm-vein images from EM emanations to spoof authentication systems. Despite promising results, two primary limitations remain. First, our proof-of-concept utilizes low-cost hardware to highlight accessibility, which currently restricts the attack to short ranges. However, since PVSPY is receiver-agnostic, the attack distance can be significantly extended by employing advanced USRPs. Second, our model is trained on low-resolution public datasets due to the scarcity of high-fidelity data. Incorporating high-resolution training sets would naturally enhance reconstruction quality. We plan to explore these improvements in future work under appropriate IRB oversight.

8 Related Works

Biometric Authentication. Biometric authentication is ubiquitous in modern devices and access control systems. Despite this widespread deployment, recent studies reveal that major biometric modalities remain vulnerable to spoofing and eavesdropping. For instance, facial recognition systems can be compromised by reconstructing 3D features from single 2D photos [38]. Concurrently, researchers are exploring novel biometric modalities to enhance security. Examples include BioFACE-3D [36], which reconstructs 3D facial geometry via ear-mounted sensors, and ECHOEXPRESS [12], which utilizes sensors in smart glasses.

Electromagnetic Side Channel. There are many efforts towards exploiting electromagnetic side channels in mobile devices to uncover user privacy. For instance, an attacker can leverage the EM emanations to infer cryptographic keys [10], unlocking passcodes [17], neural network architectures [25], and hardware or software usage in other embedded systems [20, 21]. Recent studies have shown that adversary can exploit EM emanations to reconstruct images in embedded cameras [23], fingerprints in smartphone sensors [27], and iris information in access control systems [19]. PVSPY exploits subtle analog electromagnetic emanations to reconstruct 2D palm-vein biometrics and successfully spoof COTS systems. Our work

demonstrates that deep learning can effectively recover sensitive information from weak, noisy analog signals, challenging the assumption that low signal amplitude guarantees security. Furthermore, we identify this as a fundamental hardware vulnerability likely inherent to all sensors utilizing ADCs, regardless of the sensing modality.

We note a concurrent work, EMPALM [37], which also explores the feasibility of reconstructing palm-vein biometrics via EM emanations. However, PVSPY distinguishes itself in three key aspects: (i) *End-to-End Implementation*: We demonstrate physical spoofing attacks on five commercial devices in real-world scenarios, whereas EMPALM is limited to digital reconstruction without physical validation. (ii) *Broader Scope*: PVSPY effectively targets two authentication modalities, unlike EMPALM which focuses solely on palm veins. (iii) *Practicality*: PVSPY utilizes a low-cost, portable setup that ensures high stealthiness. In contrast, EMPALM relies on bulky and expensive equipment (e.g., USRP) to capture EM leakage.

9 Conclusion

In this paper, we present PVSPY, a novel side channel attack that exploits electromagnetic emanations to spoof palm vein scanners. Using a small device placed near the target, our framework captures emanations during the authentication process and reconstructs palm vein images for physical spoofing. Extensive evaluations demonstrate that PVSPY effectively recovers biometric data, achieving a top 1 success rate of 26.7% and a top 3 success rate of 63.3% in end to end spoofing attacks across various scanners. To the best of our knowledge, this is the first work to identify and experimentally validate an electromagnetic side channel vulnerability in palm vein authentication systems.

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